Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8

Week 7

Lecture 13

13.1 Shor's algorithm

13.2 The Fourier transform

13.3 Measurement and post-processing

13.4 Summary

Lecture 14

14.1 Post-quantum cryptography

14.2 Quantum Key Distribution (QKD)

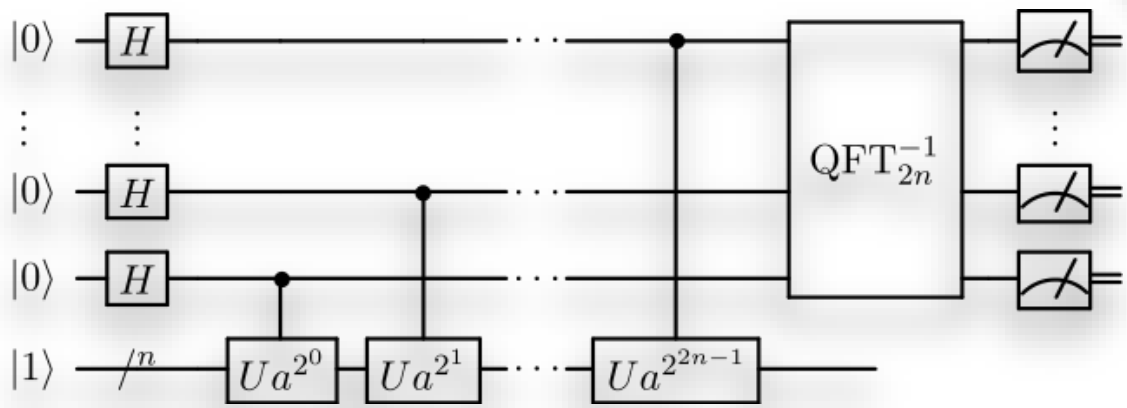
Practice class 7

Shor and QKD

Recap: Shor's algorithm

- Efficient quantum algorithms for **factoring** semiprime numbers
- Best known classical algorithm is number field sieve (exponential in bit-length).
- Underpins the RSA cryptosystem
- Hidden Subgroup Problems (e.g. Discrete logarithm) similar.

Peter Shor



Shor, Proc 35th Ann Symp of Comp Sci, 26, (1995)

14.1 Post-quantum cryptography

This section is for Information only - not examinable

References:

NISTIR 8105: Report on Post-Quantum Cryptography, available from https://csrc.nist.gov/publications/nistir

NIST: National Institute of Standards and Technology

Need for Post-Quantum Cryptography

Public key Cryptography is ubiquitous- it is a backbone of network based modern communication infrastructure.

Main tools are Encryption, Signature and key exchange protocols.

Most of these modern public key algorithms depend on Integer Factorization and discrete logarithm problems.

Shor's algorithm solves the above problems efficiently on a quantum computer.

However, quantum computing capability is currently limited.

But it is only a matter of time before we have a large quantum computer capable of solving the problems that are currently used in cryptography.

Let's now examine the impact in more detail.

Impact of Quantum Computing: NIST

Table 1 - Impact of Quantum Computing on Common Cryptographic Algorithms

Cryptographic Algorithm	Type	Purpose	Impact from large-scale quantum computer
AES	Symmetric key	Encryption	Larger key sizes needed
SHA-2, SHA-3	----------	Hash functions	Larger output needed
RSA	Public key	Signatures, key establishment	No longer secure
ECDSA, ECDH (Elliptic Curve Cryptography)	Public key	Signatures, key exchange	No longer secure
DSA (Finite Field Cryptography)	Public key	Signatures, key exchange	No longer secure

We only looked at RSA in last week's lecture. Other schemes mentioned above are equally important in practice;

Some even predict that the development of a large scale Quantum Computer is merely a significant engineering challenge; Thus it is important to find algorithms that are secure against both Quantum and Classical adversaries

Table 1: Taken from NIST report NISTIR 8105 from https://csrc.nist.gov/publications/nistir

Complexity Classes P and NP

Class P: We say that an algorithm solves a problem in polynomial time if its worst case complexity is $O(p(n))$, where $p(n)$ is a polynomial of the input size n. *Class P* is a set of all algorithms that has polynomial time complexity.

Example: Multiplication, addition and GCD between any two numbers,

Nondeterministic algorithms: Algorithms that have two stages; a guessing (oracle) stage giving a solution and a subsequent verification stage that verifies the guessed results.

If the **verification stage** of the Nondeterministic algorithm has polynomial time complexity then the algorithm belongs to *Class NP*.

Example: Factoring a large integer n; this problem is in NP, because once the factors are determined, they are the correct result if the multiplication of all the factors lead to n.

Clearly, $P \subset NP$ But, $P \; ?= NP$ An important open problem

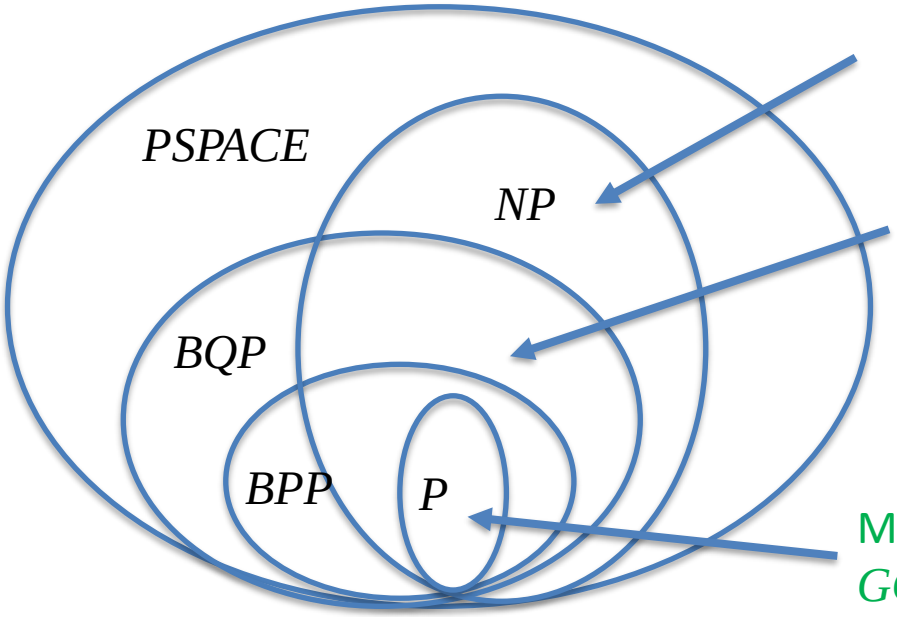
The concept can be extended to randomized algorithms; *Class BPP* is the class of all problems which can be solved by randomized polynomial time algorithms with bounded error. This is a rough definition.

Similarly we can define *Class BQP* for problems solvable by a quantum computer in polynomial time

Next we will consider more general classes and their relations to post quantum cryptography.

Complexity Classes

Where to look for candidates for post quantum cryptography?



3-Coloring, SAT, etc

Integer Factorization
Discrete Logarithm etc

Multiplication, Sorting, GCD etc

P: Polynomial Time
BPP: Bounded-Error Probabilistic P Time
BQP: Bounded-Error Quantum P Time
NP: Nondeterministic P time

BQP is Tractable on Quantum computers

Fig: The known relationships between some of the most important complexity classes.

Need hard problems above BQP Class in order to resist quantum attacks

Big challenge for quantum computing: find new problems in BQP but that are not in BPP. If you find one you will be famous!

Figure taken from: An Introduction to Quantum computing by P. Kaye et.al., OXFORD Press

NIST Suite of Public Key Post-Quantum Algorithms

There is a need to look for public key algorithms which are believed to be resistant against both classical and quantum computers.

This is a big area of research currently followed in cryptographic circles.

Some important class of algorithms currently considered by NIST are:

1. Lattice-based cryptography
2. Code-based cryptography
3. Multivariate polynomial cryptography
4. Hash-based signatures
5. Other methods based on variants of curves defined over finite fields.

See more details from: https://csrc.nist.gov/Projects/post-quantum-cryptography/

Symmetric key algorithms also affected by the developments in quantum computing, but the effect is relatively less compared to their public key counterparts; A general remedy is to double the key size. Grover's algorithm will be discussed later in the course.

Comparison of key sizes

	Signatures	Key Exchange	Fast?
Elliptic Curves	64 bytes	32 bytes	✓
Lattices	2.7kb	1 kb	✓
Isogenies	✗	330 bytes	✗
Codes	✗	1 mb	✓
Hash functions	41 kb	✗	✓

Table 1: Comparison of classical ECC vs post-quantum schemes submitted to NIST

This comparison is only for schemes submitted to NIST

This is a very active area in Cryptography currently, please follow the research from the NIST resources.

Table from: https://blog.trailofbits.com/2018/10/22/a-guide-to-post-quantum-cryptography/

14.2 Quantum Key Distribution

Back to material that are examinable

Key Distribution

Alice would like to establish a secret key with Bob.

Alice

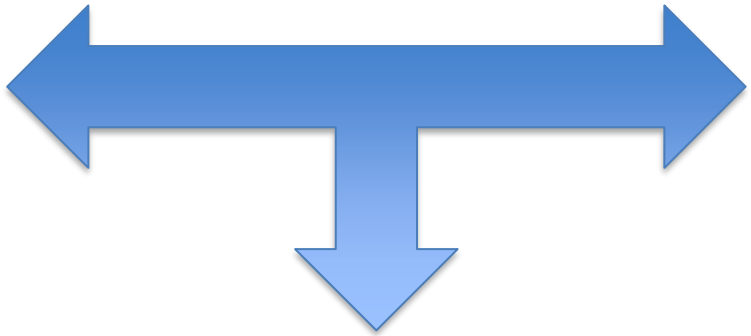
Secret key
0111011000111

Bob

Alice and Bob would like to establish a secret key. If they had such a key they could use it to communicate securely. The problem is Alice and Bob are both stuck at home due to COVID-19 so they can't get together to establish a secret key in private.

Eve, the eavesdropper

Alice



Bob

Eve

But if Alice tries to send Bob a key (or vice-versa) when their communication is not secure, then an eavesdropper (called Eve) can listen in an learn their key.

Story so far

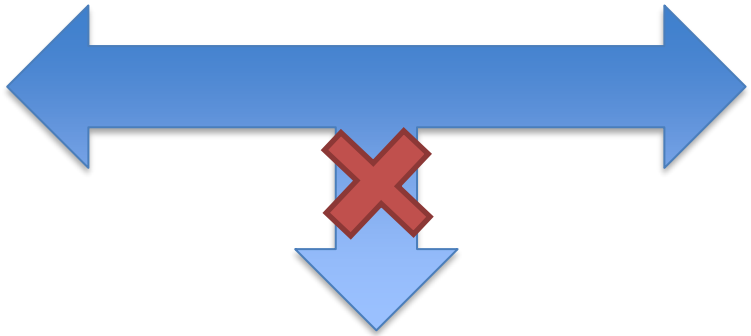
- Last lecture we demonstrated how a Quantum algorithm breaks RSA, a popular public key cryptography scheme. Note that RSA is the first efficient public key scheme where two users who have not met before is able to arrive at a common secret on a public communication link by using only public information.

- Here we will see how quantum protocol provides a similar result, i.e., concrete secret key exchange scheme between two communication entities.

- So, what was taken away Quantum gives back in another form.

Quantum Key Distribution

Alice



Bob

Eve

Quantum Key Distribution is a way that Alice and Bob can establish a secret key, and know that an eavesdropper, Eve, has *not listened in* as they do that.

Stages of BB84 Quantum Key Distribution

Quantum Key distribution follows several steps.

1) Alice chooses to encode a qubit in a random basis
2) Alice sends qubits to Bob
3) Bob chooses to measure in a random basis
4) Alice and Bob announce their basis. They throw out qubits where their bases differ.
5) On a subset of qubits, Alice and Bob compare results
   - If they disagree on these qubits, there might be an eavesdropper
   - If they agree, they use the state of the remaining qubits as the key.

Let's now examine each step in more detail.

Alice encodes in a random basis

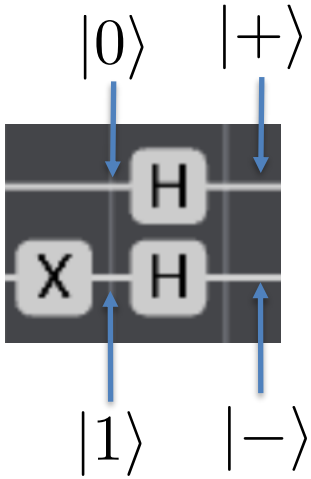
What do we mean by a random basis? Alice either encodes using the z-basis or the x-basis:

Binary Data	z-basis	x-basis
0	$\|0\rangle$	$\|+\rangle$
1	$\|1\rangle$	$\|-\rangle$

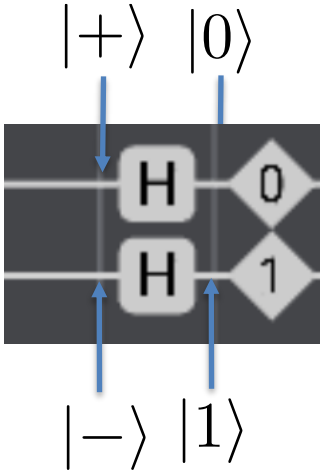
where $|+\rangle = \dfrac{|0\rangle + |1\rangle}{\sqrt{2}}$ $\quad |-\rangle = \dfrac{|0\rangle - |1\rangle}{\sqrt{2}}$

Sending and receiving in the x- basis

To change between x- and z- bases, just add a Hadamard gate:



Sends qubit

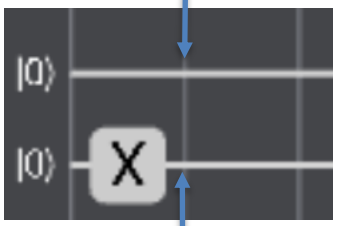


Alice sending in x-basis

Bob receiving in x-basis

Sending and receiving in the z- basis

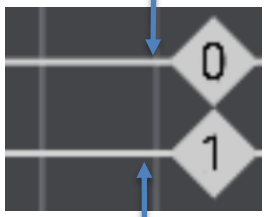
$|0\rangle$



$|1\rangle$

Sends qubit

$|0\rangle$

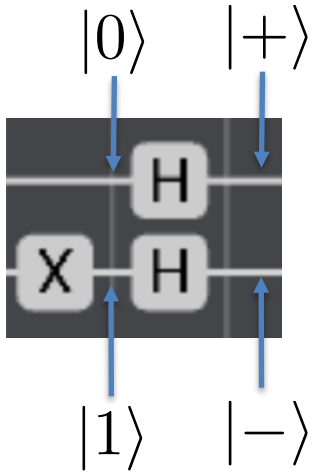


$|1\rangle$

Alice sending in z-basis

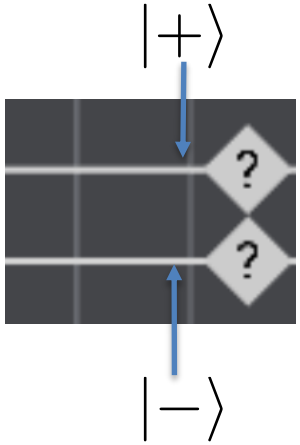
Bob receiving in z-basis

If they don't agree on the basis…



Alice sending in x-basis

Sends qubit

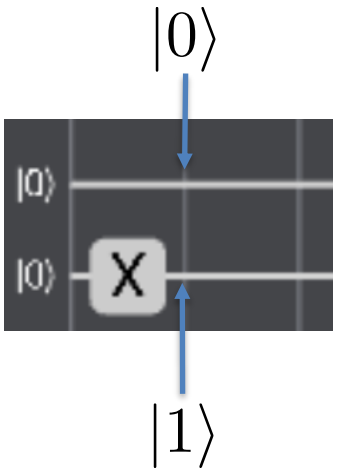


Bob receiving in z-basis

Measurement results: 50/50%

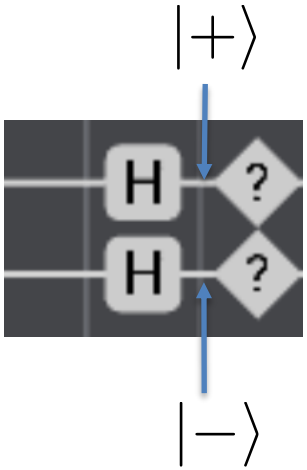
$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$



Alice sending in z-basis

Sends qubit



Bob receiving in x-basis

Measurement results: 50/50%

Measurement of |+> and |-> states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

50%

Measure "0" and state collapses to:

$$|0\rangle$$

50%

Measure "1" and state collapses to:

$$|1\rangle$$

Measurement of |+> and |-> states

If Bob measures in the "wrong" basis, then his result is 50/50%.

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

50%

Measure "0" and state collapses to:

$$|0\rangle$$

50%

Measure "1" and state collapses to:

$$|1\rangle$$

In quantum mechanics **measurement disturbs the state**.

Discard when bases disagree.

If Alice and Bob **use the same basis** (for transmission and measurement), then the bit Alice transmitted is what Bob received.

These bits can be used as part of the key.

If Alice and Bob **use different bases** (transmit in one basis, and measure in the other) then what is received is random.

These bits should be discarded.

After qubits have been distributed and measured

Alice announces what basis she used to transmit each qubit.

Discard if they disagree

Keep if they agree

Bob announces what basis he used to measure each qubit.

Example of first few qubits sent

Alice Sends	Alice Basis	State Alice Sends	Bob's Basis	Bob Measures	Keep this bit?	Bit for secret key
0	x	|+>	x	0	✓	0
1	z	|1>	x	0	✗	-
1	x	|->	z	1	✗	-
0	z	|0>	z	0	✓	0

What about the eavesdropper?

Eve, the eavesdropper, can listen in but has a *problem*:

- She doesn't know which basis to measure in, so she has to guess, randomly choosing a basis to measure in.
- If she chooses correctly, she can send on the state unchanged.
- If she chooses wrongly, she gets a random result, and sends on the state in the **wrong basis**. Note that she *cannot send on the original state*, because it has collapsed when she measured it.

In quantum mechanics measurement disturbs (by collapsing) the state you are trying to measure.

- Alice and Bob can detect the presence of Eve, they choose a subset of their qubits where they used the same basis, and check the bits really are the same:
  - Without an eavesdropper, their bits be the same.
  - With an eavesdropper, 1/4 will be different.

Detecting an Eavesdropper

Alice Sends	Alice Basis	Alice Sends	Eve's Basis	Eve measures	Eve sends	Bob's Basis	Bob measures	A/B Same basis?	Do their bits match
0	x	|+>	x	0	|+>	x	0	✓	✓
1	z	|1>	z	1	|1>	x	0	✗	-
1	x	|->	z	1	|1>	z	1	✗	-
0	z	|0>	x	0	|+>	z	1	✓	✗

There is an eavesdropper on the line!

Only some qubits are used to detect an eavesdropper. The remaining qubits are used to construct the **shared secret key**.

Stages of BB84 Quantum Key Distribution (Revisited)

Quantum Key distribution follows several steps.

1) Alice chooses to encode a qubit in a random basis
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Security of QKD Systems

QKD systems have inherent information-theoretic security based on quantum mechanical principles.

A typical cryptographic scenario: Alice and Bob can use QKD systems to obtain string of key bits (r) by BB84 protocol.

- Any eavesdropper naturally introduces noise to the transmission, the end users can estimate maximum amount of information gained by the eavesdropper.

Another process called privacy amplification (not discussed in this lecture) can be applied to compress (r) to a final secret key (k) which is completely oblivious to the eavesdropper.

Usual PA processes are based on applying hash functions

Quantum Hacking and other attacks

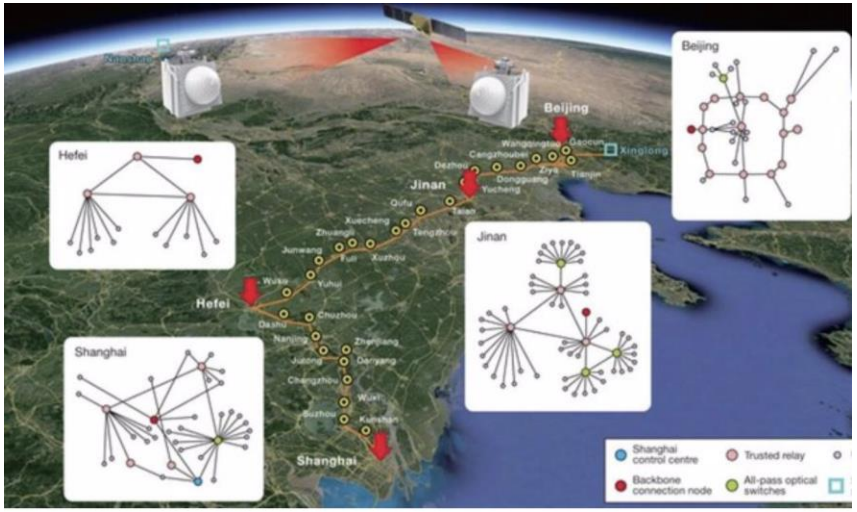
Are there any other attacks on QKD systems?

Any attack must base it on the quantum implementation's deviations with its ideal model.

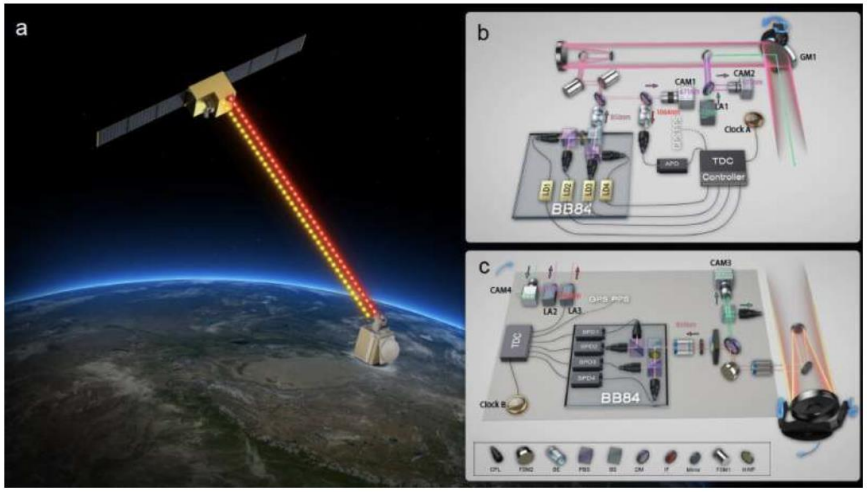
- If the gap is less than threshold, the size of the key is related to the amount of potentially leaked information which helps to determine the parameters of compression in the PA techniques.
- If the gap is greater than a threshold, end users may not realize a secure key.

There is this familiar denial of service (DoS) attack. Counter strategy is to introduce alternative channels for acquiring keys.

QKD is fast becoming a reality



QKD distribution network built over 4600km in China connecting Shanghai and Beijing.*



Next steps: satellite based QKD, with existing technology demonstrating free-space quantum communication over the atmospheric distances required.**

* Ren, JG., Xu, P., Yong, HL. *et al.* Ground-to-satellite quantum teleportation. *Nature* **549,** 70–73 (2017). https://doi.org/10.1038/nature23675

** Liao, SK., Cai, WQ., Liu, WY. *et al.* Satellite-to-ground quantum key distribution. *Nature* **549,** 43–47 (2017). https://doi.org/10.1038/nature23655

Week 7

Lecture 13

13.1 Shor's algorithm

13.2 The Fourier transform

13.3 Measurement and post-processing

13.4 Summary

Lecture 14

14.1 Post-quantum cryptography

14.2 Quantum Key Distribution (QKD)

Practice class 7

Shor and QKD